1. Given an array A[ ] of positive integers of size N, where each value represents the number of chocolates in a packet. Each packet can have a variable number of chocolates. There are M students, the task is to distribute chocolate packets among M students such that:

- Each student gets exactly one packet.
- The difference between maximum number of chocolates given to a student and minimum number of
- chocolates given to a student is minimum.

Sample Input	Sample Output
N = 7, M = 5 A = [3, 4, 9, 56, 7, 9, 12]	6
N = 7, M = 3 A = [7, 3, 2, 4, 9, 12, 56]	2

Explanation #1: The minimum difference between maximum chocolates and minimum chocolates is
9 - 3 = 6 by choosing following M packets: {3, 4, 9, 7, 9}.
Explanation #2: The minimum difference between maximum chocolates and minimum chocolates is
4 - 2 = 2 by choosing following M packets: {3, 2, 4}.
HINT: It said MINIMUM. What common pattern does the greedy follow at the start? Find the pattern in the explanations, and you will find your answer easily.

2. You are a university administrator responsible for distributing scholarships to students. However, each scholarship can only be awarded to one student.
Each student i has a minimum funding requirement r[i], which is the minimum scholarship amount they need to cover their tuition and expenses. Each scholarship j has a value v[j].
If v[j] >= r[i], we can assign scholarship j to student i, and the student will accept it.
Your goal is to maximize the number of students who receive scholarships and output that maximum number.

Sample Input	Sample Output
r = [10000, 20000, 30000] v = [10000, 10000]	1
r = [15000, 25000] v = [20000, 30000, 40000]	2

Explanation #1:
- Student A needs $10,000 minimum
- Student B needs $20,000 minimum
- Student C needs $30,000 minimum
- You have two scholarships, both worth $10,000

Only Student A can be satisfied. Students B and C require more than any available scholarship can
provide.

Explanation #2:
- Student A needs $15,000, can receive the $20,000 scholarship
- Student B needs $25,000, can receive the $30,000 scholarship
- The $40,000 scholarship remains unused

Both students are satisfied.

3. You are shopping in a store with N products, each having a certain price (possibly with decimals). The store gives you K discount coupons. Each coupon can only be used on one product and each product cannot receive more than one coupon. If a coupon is applied to a product, its price is reduced by 30. That means if a product price = P, then after applying the coupon, its new price = P × 0.7. Example: If the initial price is 60, then after applying discount the price will be 60 * 0.7 = 42 Coupons should be applied in such a way that the total cost of all products is minimized. Write a program using a greedy algorithm to solve this problem.
Input: The first line contains two integers N and K. The next line contains N double values, price of N products.
Output: A single double value representing the minimum total cost.

Sample Input	Sample Output
5 2 20.50 50.00 30.75 10.20 60.30	138.66
4 1 100.99 90.50 80.20 70.75	312.14

Explanation: For Sample Input 1, the coupon will be applied to product 2 and 5. So, new prices = [20.50, 35.00,
30.75, 10.20, 42.21]. Total = 138.66
For Sample Input 2, the coupon will be applied to product 1. So, new prices = [70.69, 90.50, 80.20, 70.75]
Total = 312.14

4. Jarif got stuck on an island. There is only one shop on this island, and it is open on all days of the week except for Sunday. Consider following constraints:
N – The maximum unit of food you can buy each day.
S – Number of days you are required to survive.
M – Unit of food required each day to survive.
Currently, it's Monday, and he needs to survive for the next 'S' days.
Find the minimum number of days on which you need to buy food from the shop so that he can survive the next 'S' days or determine that it isn't possible to survive.

Example 1:
Input: S = 10, N = 16, M = 2
Output: 2
Explanation: One possible solution is to buy a box on the first day (Monday), it's sufficient to eat from this box up to 8th day (Monday) inclusive. Now, on the 9th day (Tuesday), you buy another box and use the chocolates in it to survive the 9th and 10th day.

Example 2:
Input: S = 10, N = 20, M = 30
Output: -1
Explanation: She can't survive even if she buy food because the maximum number of units she can buy in 1 day is less the required food for 1 day.
Primary Task: Create a function minimumDays() which takes S, N, and M as input parameters and returns the minimum number of days Ishika needs to buy food. Otherwise, returns -1 if she cannot survive.
Constraints:
$1 \leq N, S \leq 50$
$1 \leq M \leq 30$

5. Following items are available in a grocery shop:

- ➢ 12 kilogram rice grain which costs 840 taka
- ➢ 10 kilogram salt which costs 870 taka
- ➢ 8 kilogram saffron powder which costs 2000 taka and
- ➢ 5 kilogram sugar which costs 500 taka

A group of thieves (Thief 1, Thief 2, … Thief M) have come to steal from that shop, **each with a knapsack of capacity 9 kg**. The thieves are entering in serial, Thief 2 enters after Thief 1 is done with stealing, Thief 3 enters after Thief 2 is done with stealing and so on. Since each thief wants to maximize his/her profit, **how many thieves** will be needed in the group to empty the grocery shop and **what are the items** that each of those thieves carry? Also print each thief's profit.

6. You are given the arrival and the departure times of eight trains for a railway platform, and each one is in the format: [arrival time, departure time). Only one train can use the platform at a time. There must be at least 1 unit time gap between 2 consecutive trains. Suppose that you have got the following train-use requests for the next day.

{ [8, 12), [6, 9), [11, 14), [2, 7), [1, 7), [12, 20), [7, 12) , [13, 19) }

Find the **maximum number of trains** that can use the platform without any collision by using *earliest departure time*.

7. https://elms.uiu.ac.bd/mod/resource/view.php?id=88581 - all problems given in the pdf.